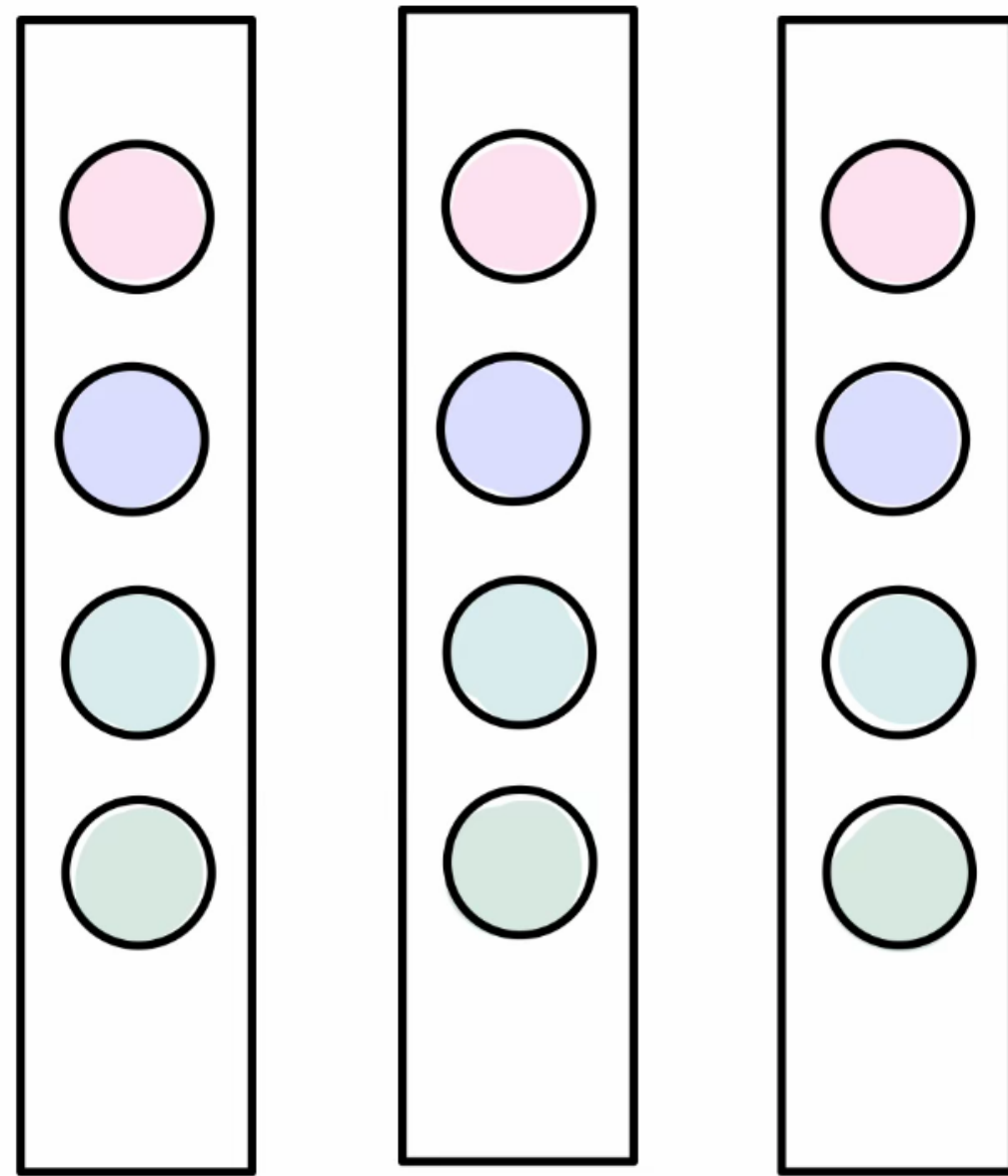


Duplicated Structured Representations

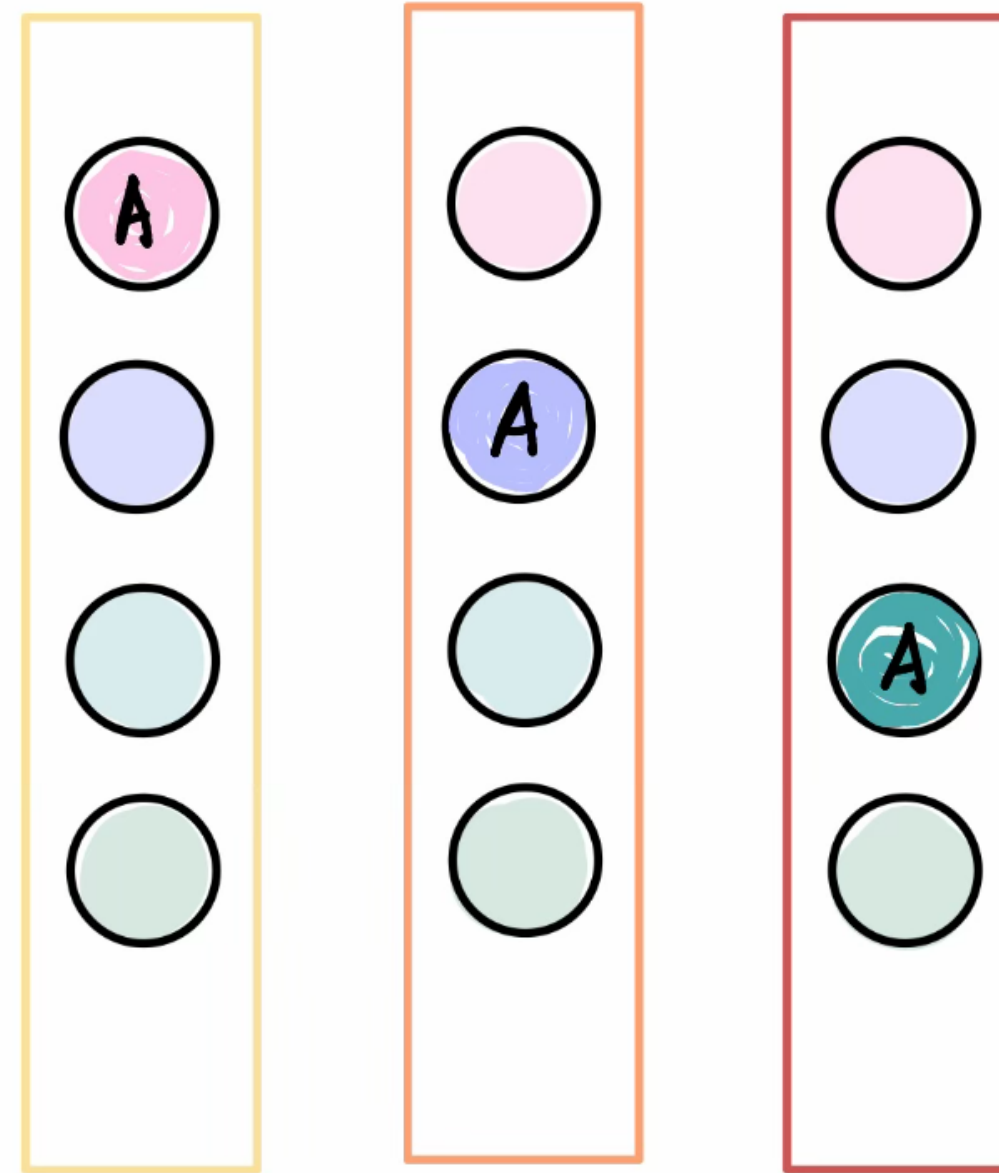
Explained

Ruby Tseng, July 21 2026

Duplicated



duplicated

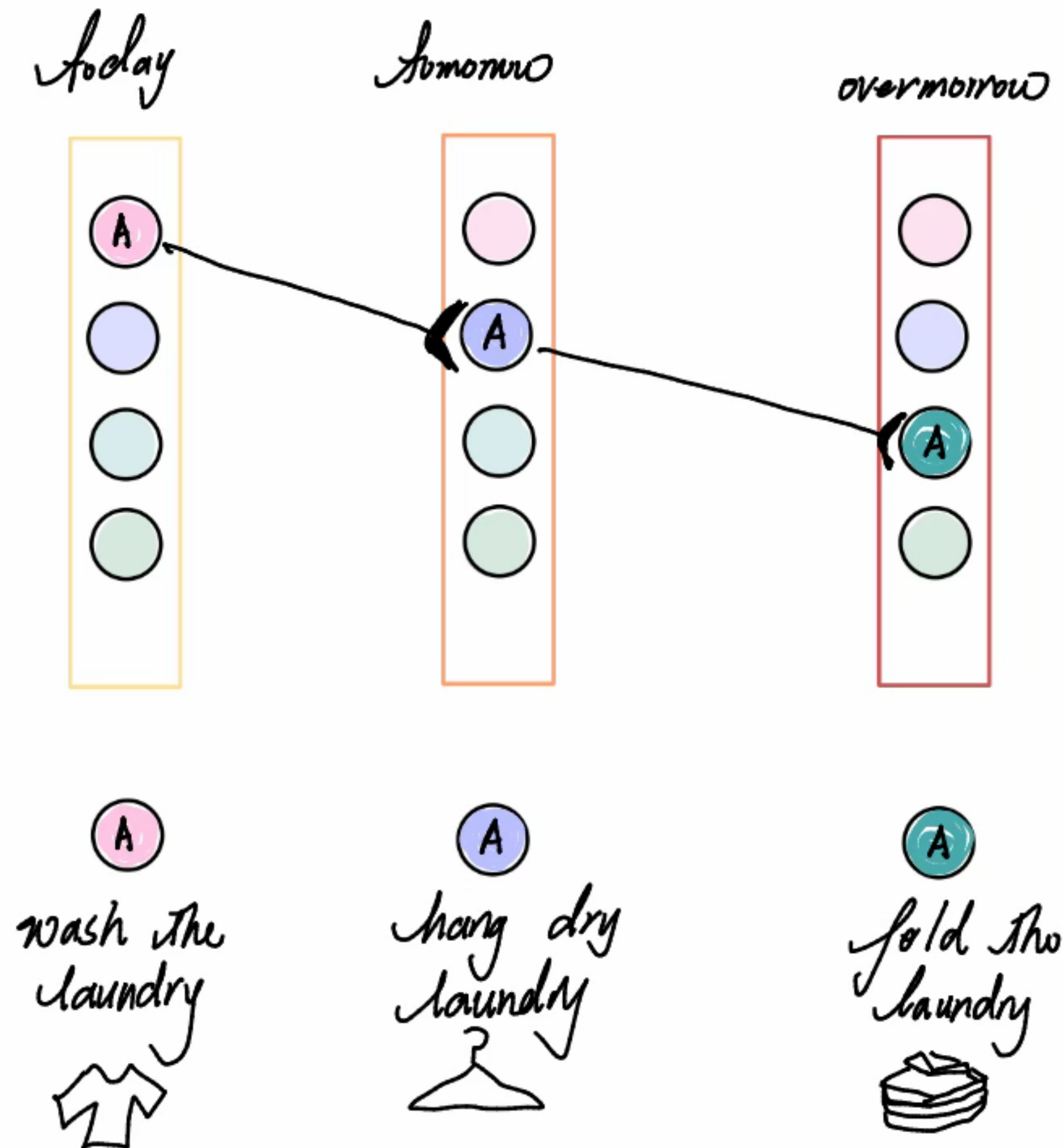


same action, different context

- The same action  is duplicated by **different** neuronal populations in **different** contexts
- Each action  is conjugated with two features action + context
- Each of the possible action subspaces is duplicated



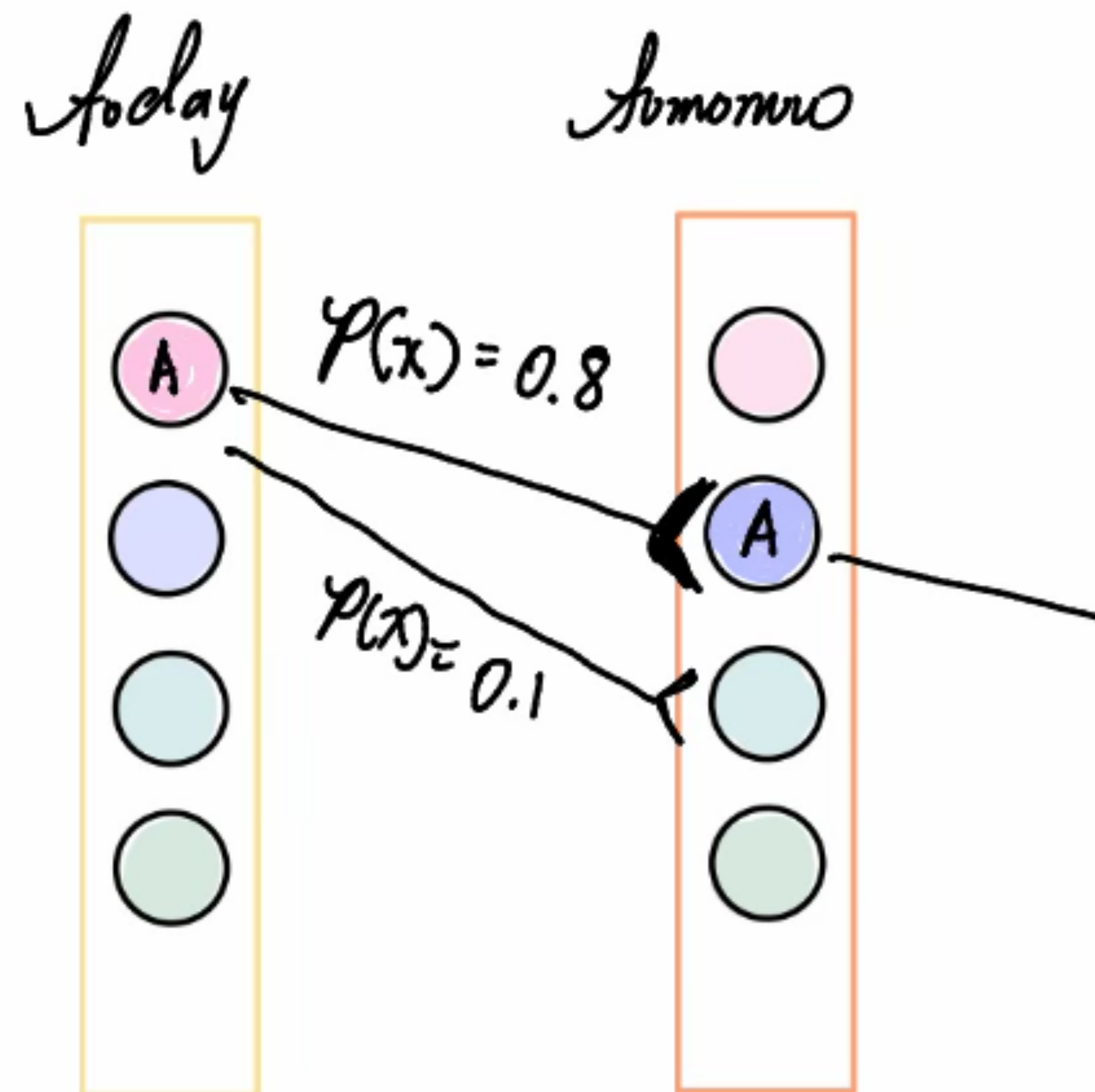
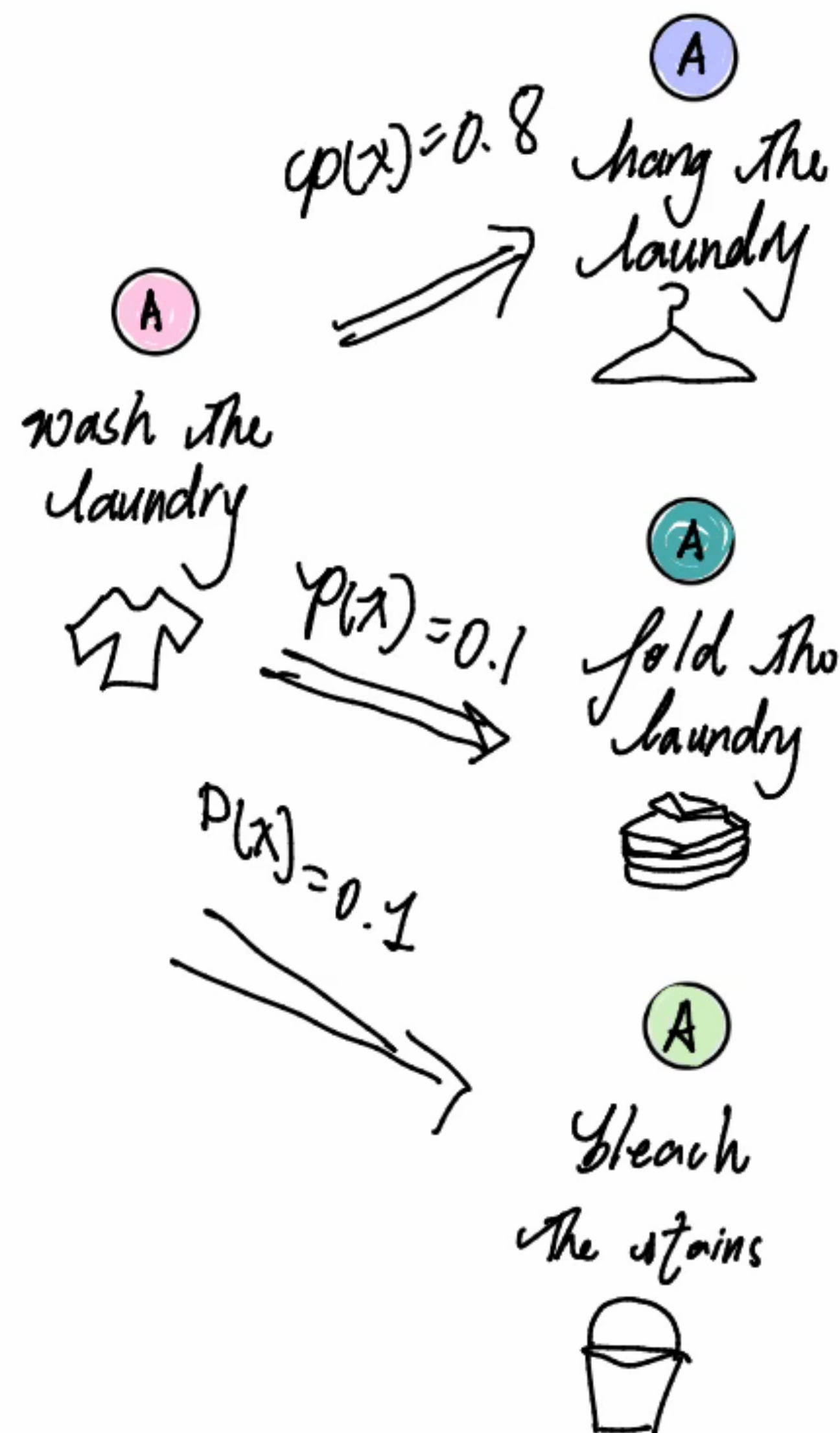
Structured



- Duplicated representations embed and are constrained by structured knowledge of world

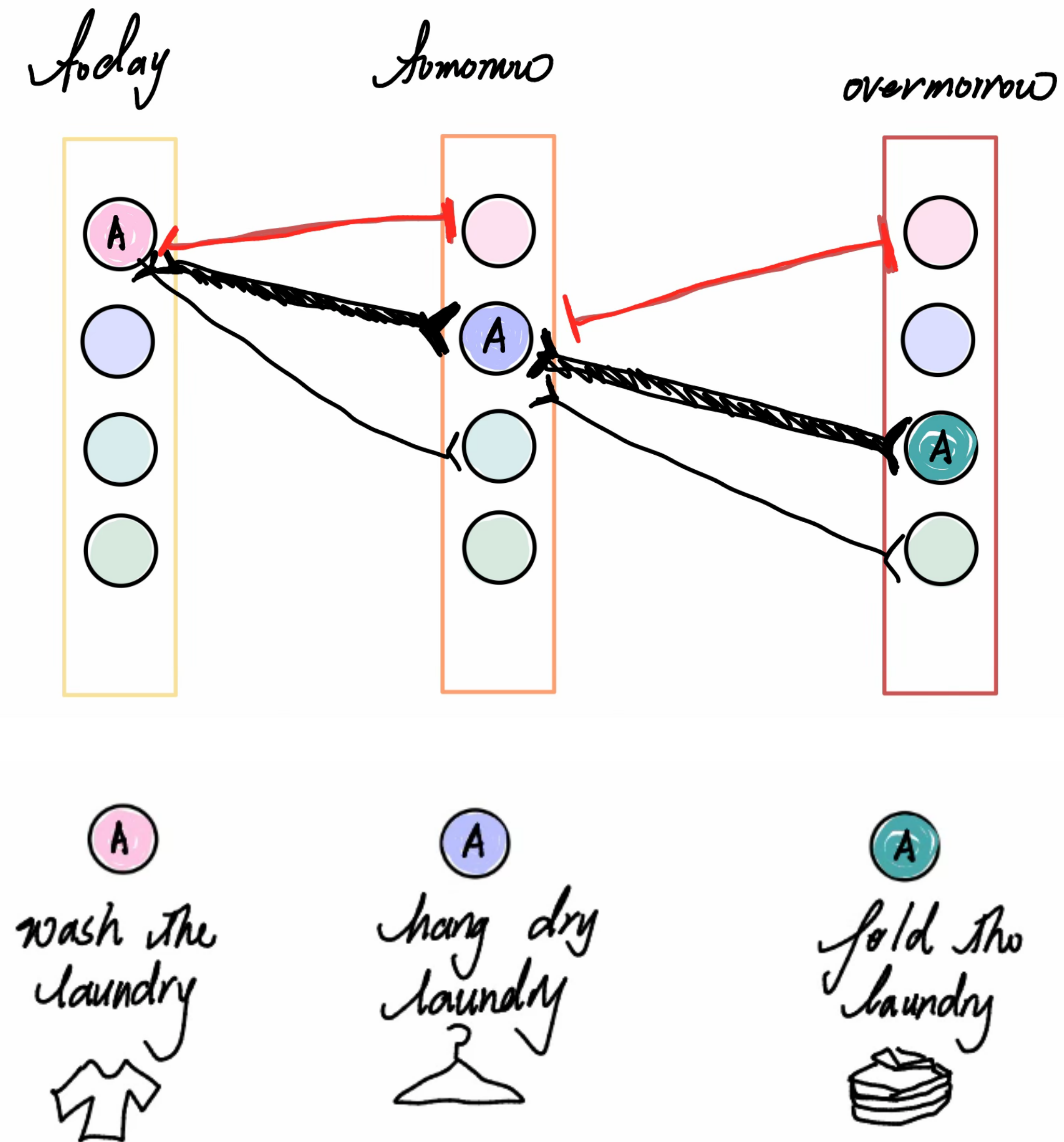
Structured

- Duplicated representations are structured by **statistical relationships**



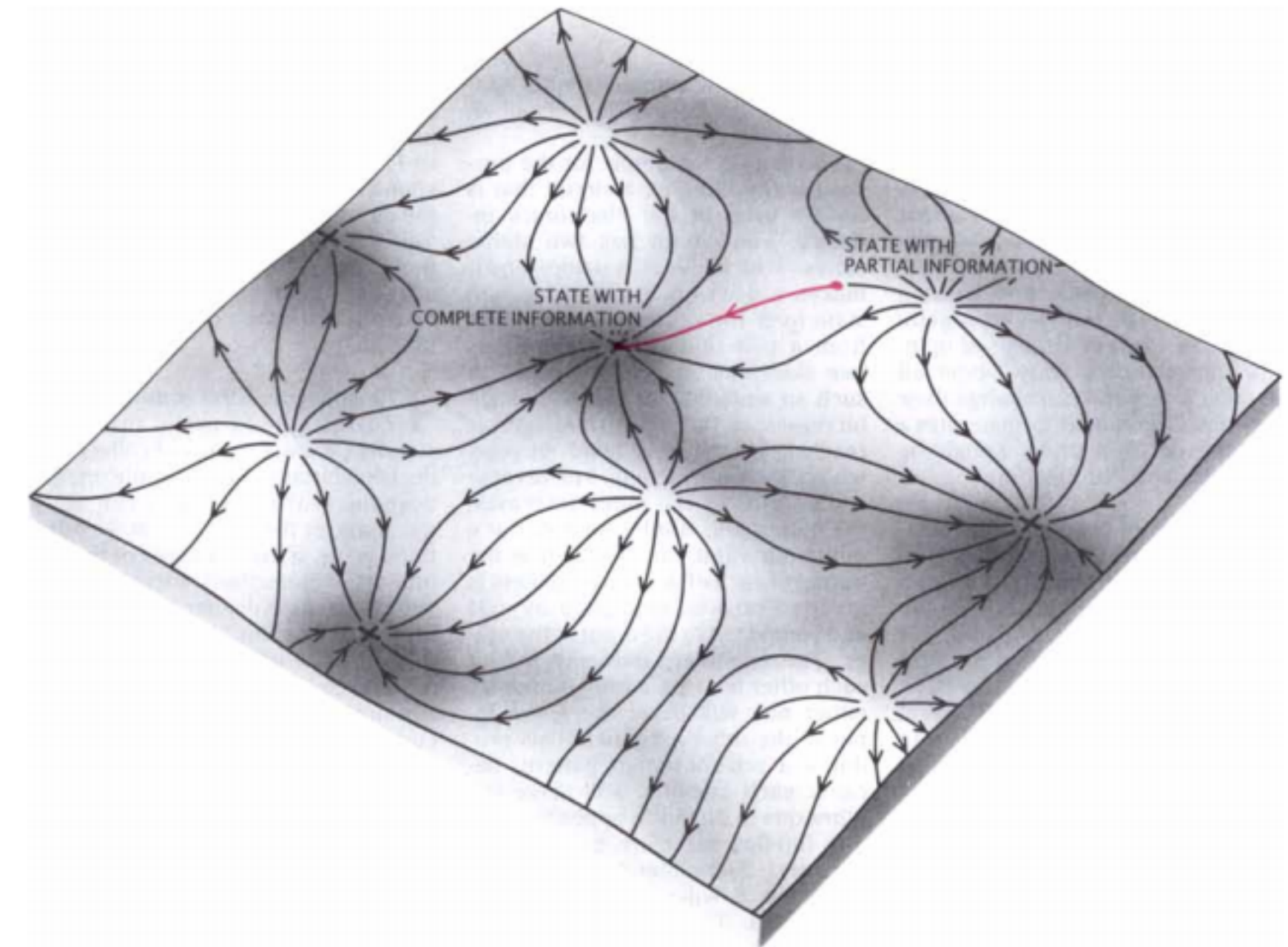
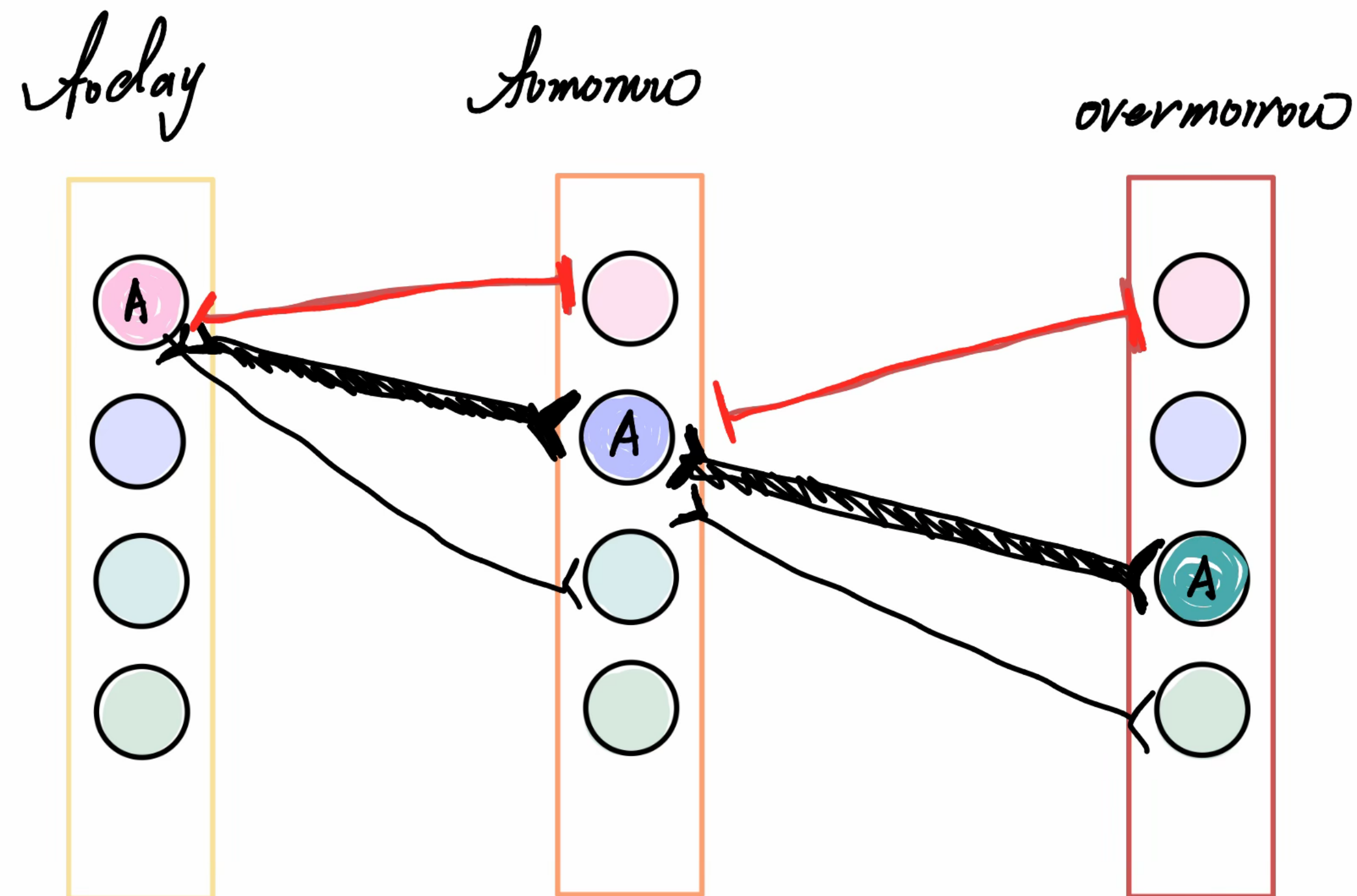
Structured

- Bidirectionally of synaptic connections (**recurrent connections**)
- Allows for planning of actions based on future targeted actions
- e.g I want to do laundry over morrow, what actions do I need to take the days before that?



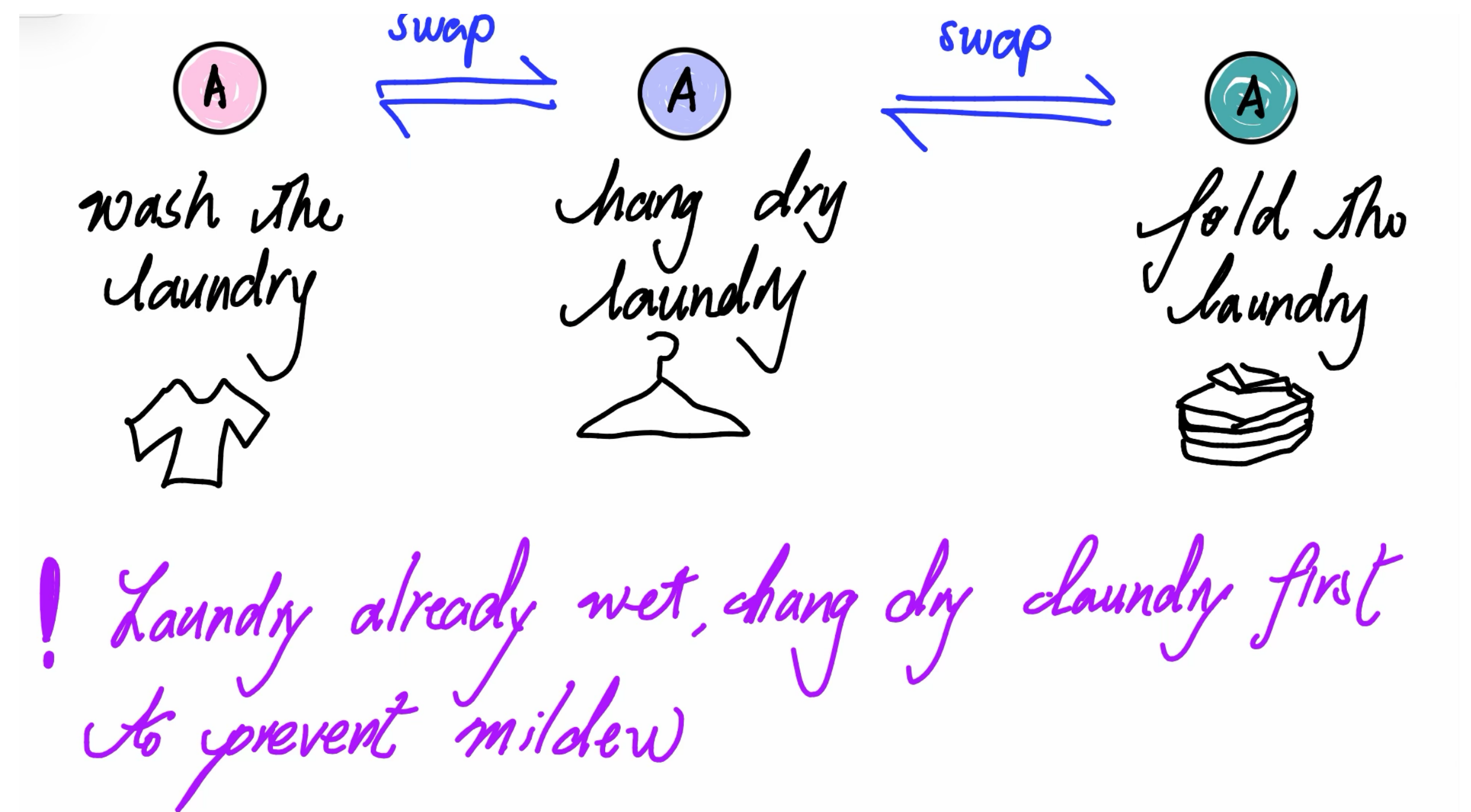
Structured

- **Persistence activation** of connectivity stabilises action
- Embeds “rules” for the context
- Stabilisation become “**attractor**” states



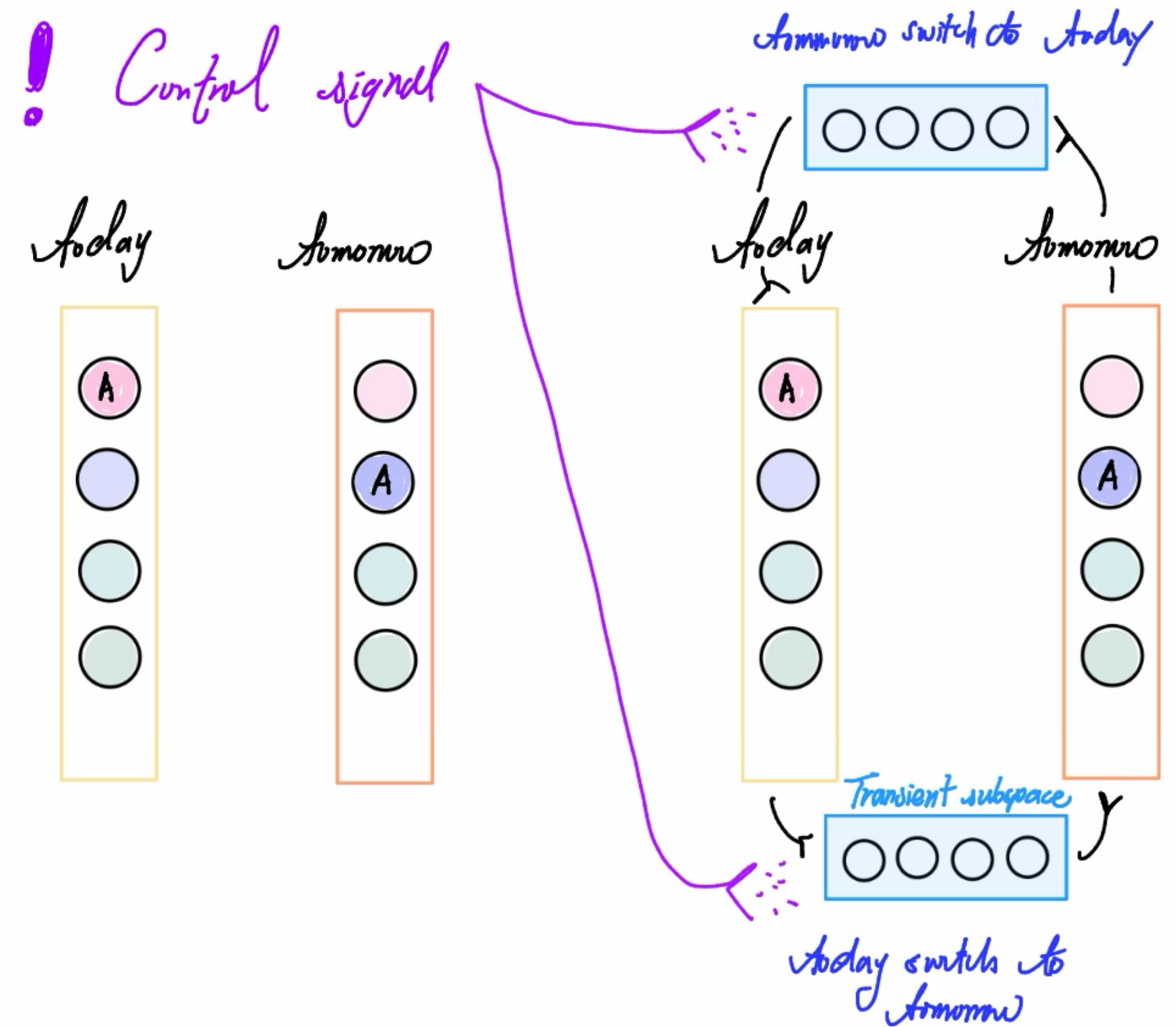
Structured

- **Gating** from external and internal signals that indicates a change of action



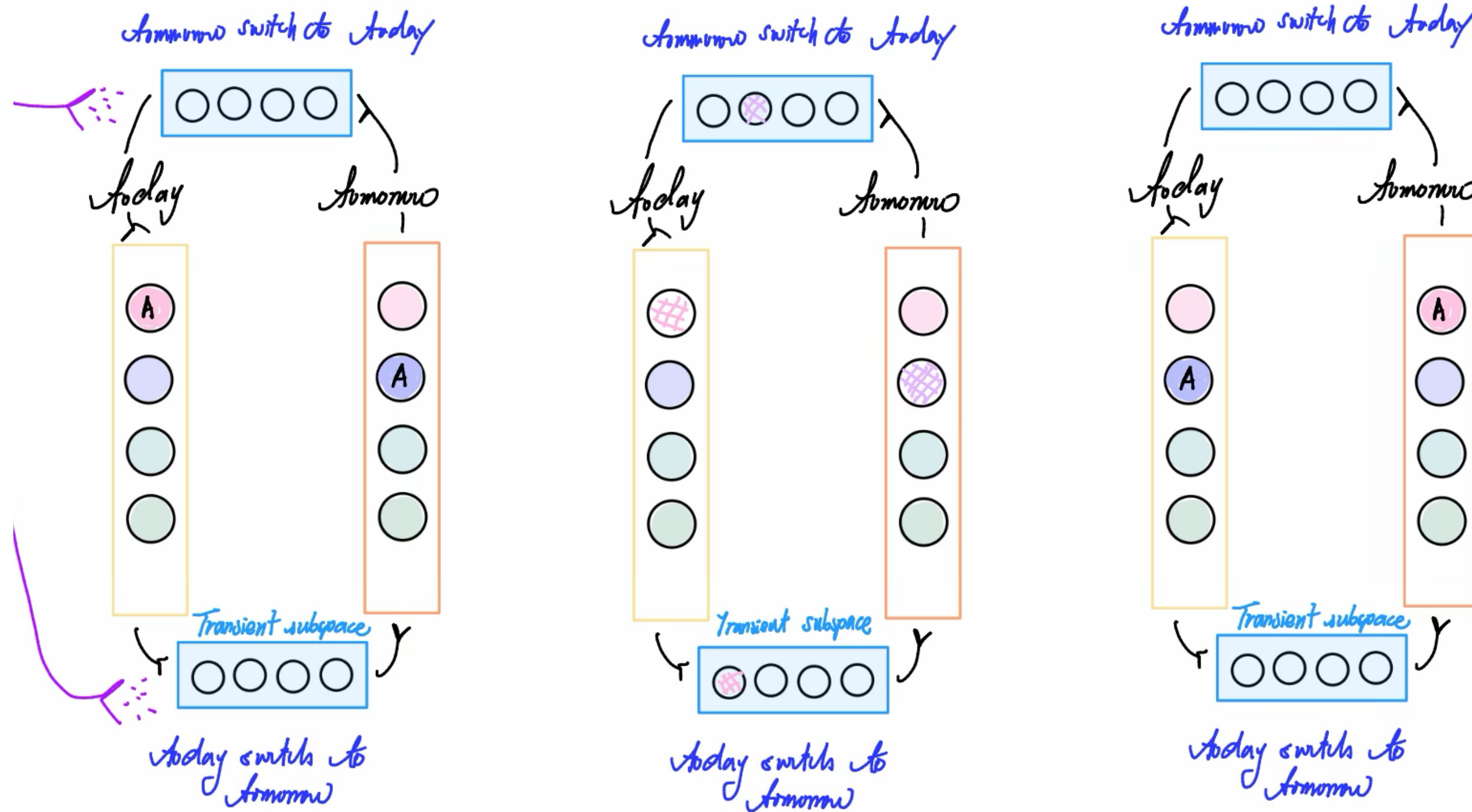
Structured

- Structural connectivity can be **gated**.
- Control signal activates transient subspace neurons by bringing them over the threshold

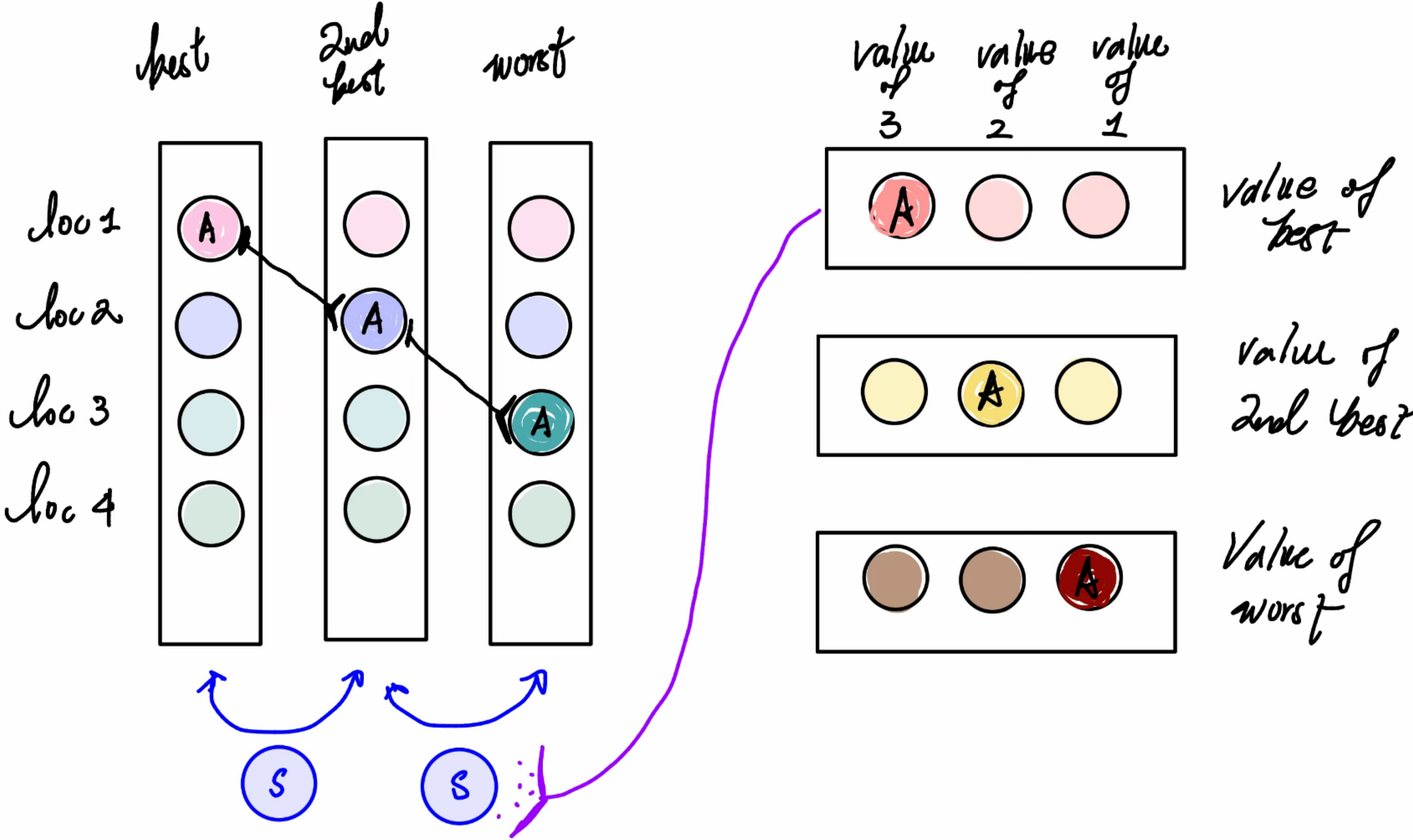


Structured

- Transient subspace temporarily holds action information as transitions occur



Meg's Task



Add data that shows this theory

Generalisability

What do DSRs offer

- Efficient linear readout
- Conjunctive coding of actions allows for generalisation across different situations
- Computational and neuronal efficiency
 - A single neuron can encode more information
- Allows for inference and planning